

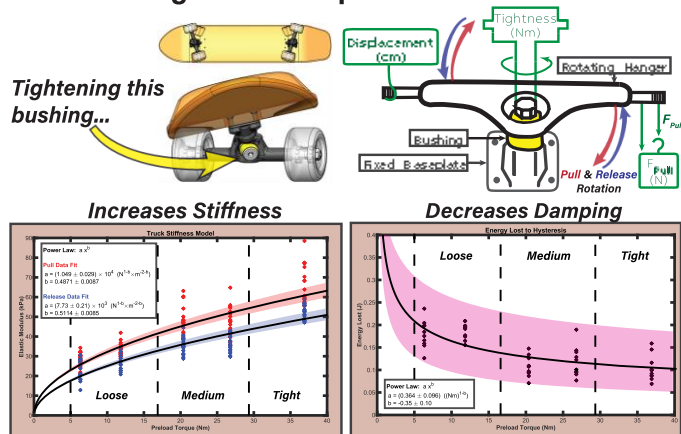
THE EFFECT OF BUSHING TIGHTNESS ON SKATEBOARD DYNAMICS

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VISUAL ABSTRACT

How Tightness Shapes Skateboard Feel



ABSTRACT

Skateboard performance depends critically on the tightness of the preloading bolts, yet even the most experienced riders struggle to precisely adjust their skateboard's trucks to their preferred stiffness and damping. To address the lack of quantitative data linking bushing preload to truck parameters, this study isolated a skateboard truck to measure force–displacement response under known preload torques. A pulling load induced a measured deflection; MATLAB analysis computed elastic moduli from beam-bending relations, separated the Pull and Release phases, and plotted energy dissipation. Results show a positive, nonlinear correlation between torque and stiffness, following an approximate square-root trend where stiffness increased by 97% across the tested preload

range. The model of energy loss quantifies the expected viscoelastic decaying hysteresis trend that the 21% offset between Pull and Release elastic moduli also validates. These findings establish a quantitative basis for skateboard tuning, connecting adjustment to board feel through the stability in stiffness and the responsiveness in hysteresis.

Keywords: Polyurethane, Elastomer, “Compressive Strength”, “Elastic Modulus” OR “Modulus of Elasticity” OR “Young’s Modulus”, “Shore A Hardness”

INTRODUCTION

Skateboarding has expanded from a subcultural mode of entertainment and expression into an Olympic sport and multi-billion-dollar industry, where equipment design and performance data carry both cultural and commercial weight [1]. As the sport's profile grows, accurate models of skateboard dynamics have become critical to an increasingly wide range of stakeholders: performance scientists evaluating equipment and technique, footwear and apparel companies analyzing forces for design, game developers seeking satisfying realism in simulations, and most importantly, the skateboarders that these groups revolve around. Yet despite this demand, information on skateboard mechanics remains sparse, as much of the research is held proprietary by companies competing to maintain small leads in research and development.

Among the components that govern each skateboard's unique performance, the compliant bushings play a central role by governing the turning response. As every skateboard requires two pairs of bushings and casting them in polyurethane is relatively inexpensive at scale, the manufacturing and distribution of the part studied in this

project constitute a considerable portion of the skateboarding market. For brands, there exists a strong incentive to reduce the barrier to entry for purchasing these individual parts to access beginner-friendly markets that historically purchase pre-assembled skateboards with pre-selected parts. These pre-assembled skateboards from big-box retailers often cost more while sacrificing quality, but small skateboard businesses struggle to gain market share due to the daunting hurdles for consumers such as selecting the correct bushing hardness and tightening it the proper amount. Analysis explicitly connecting bushing preload to mechanical stiffness would democratize previously limited information, opening the beginner market to small businesses and the sport itself for the next generation of skateboarders.

From the dedicated skateboarder's perspective, these small businesses, known as skate shops, are a matter of life and death for the community [2]. Tightknit local skateboarding communities motivate skateboarders to persist in the physically taxing sport, and skate shops provide a space for that community to congregate, grow, and provide for itself. Staffed and owned by local skateboarders, skate shops reinvest in the community by hosting events and supporting new skateboarders, as shops are incentivized by the demand a healthy community brings. The task of democratizing information for small businesses is increasingly relevant as private equity continues to seize market share with larger marketing and R&D budgets [2]. The democratization of bushing information would remove a major hurdle for new skateboarders, demystifying tacit knowledge to provide a less imposing experience for beginners making their first purchases at a local shop.

This project investigated how compressive preload, controlled by tightening the kingpin nut, affects the stiffness and energy efficiency of skateboard bushings and, by extension, the turning characteristics of a skateboard. To address this objective, an isolated truck was tested under varied loading conditions. For each trial, the kingpin was tightened to a known torque, after which a DFS force sensor applied a controlled load at a fixed location on the wheel axle. This load imposed a calculable moment on the bushings, and the resulting torsional deflection was recorded by a linear motion transducer positioned relative to the truck's axis of rotation. The data was processed into force-displacement curves, differentiated into stiffness values, and integrated into energy consumption, so stiffness and energy consumption could be plotted against tightness and fit to models. This analysis quantifies trends such as diminishing returns at higher preloads and frames

previously experiential information in ways accessible to diverse audiences, from engineers to skateboarders themselves.

Quantitative models make accessible framing possible with the dual benefits of streamlined system analysis and ease of comparison. Since stiffness is the ratio of force to displacement and a model produced in this project relates stiffness to torque, a skateboarder's weight and the known height of their trucks, which controls the maximum possible displacement, is enough information to determine what torque would achieve reasonably loose trucks for any skateboarder. This torque value can then be communicated as a multiple of an understood torque, such as the torque to open a tight jar lid, which is roughly 5 Nm, or the bushing's maximum tightness, which occurs around 40 Nm when the preloading kingpin stops rotating smoothly. By connecting mechanical preloads to turning behavior, this work contributes both a fundamental insight into the compliant mechanisms at the heart of skateboarding and a practical, democratized framework for equipment design and informed consumer choice.

BACKGROUND

Understanding how the tightening of a skateboard's trucks influences board feel requires grounding the experiment in both the mechanical structure of the truck and the material behavior of its polyurethane bushings. The goal of this section is therefore to establish the theoretical and physical context necessary to interpret the stiffness and hysteresis measurements that follow. At the system level, a skateboard truck acts as a compliant turning mechanism in which user-adjustable preload directly modifies the rotational response of the hanger. At the material level, the polyurethane bushings that govern this response exhibit nonlinear, hyper-viscoelastic behavior that departs from ideal elastic models, critically introducing energy loss through hysteresis. Together, this context provides the conceptual framework necessary to interpret the experimental results, motivating why preload-stiffness relationships and hysteresis energy losses are both expected and technically significant in skateboard dynamics.

SYSTEM DESCRIPTION

A skateboard truck serves as the connection between the deck, on which the rider stands, and the wheels, converting moments applied in the skateboard's roll direction to wheel rotation in the yaw direction through compliant bushings in each truck. The baseplate, which fastens to the deck with four 10-32 bolts, is attached to the

hanger, which holds the wheel axle, at two points: a ball joint (“pivot cup”) and a pair of compliant bushings around a $\frac{3}{8}$ ” bolt (the kingpin). The compression preload applied to these bushings is controlled by the kingpin nut, which can be tuned by the user to set the tightness of the truck. This kingpin nut is the face of a fundamental tradeoff in skateboarding, as a truck with less tightness is associated with more maneuverability while tighter trucks bring greater stability. By isolating the system of interest to only the truck itself, variables such as deck width, deck compliance, and fastening hardware can be characterized separately by future research.

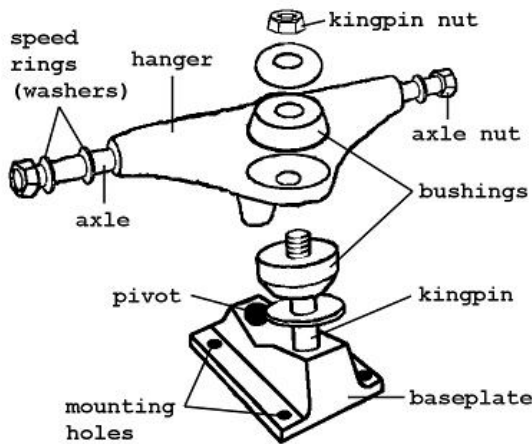


Figure 1: Exploded View of a Skateboard Truck [3]. This project considered the effect of tightening the kingpin nut on the hanger’s rotation response to imposed torques, as mediated by the bushings.

SYSTEM MODEL CONSIDERATIONS

As this project studies the properties of the bushings themselves, the system model must isolate or control other sources of variation. Conveniently, the components of trucks are already designed to minimize non-bushing contributions to the skateboard’s turning response. The Delrin pivot cup reduces friction in the ball joint, and the rigid aluminum structure ensures efficient force transfer. However, part wear and creep deformation may still affect the experimental setup over time. To mitigate these effects, the proposed tests will (1) perform an array of cyclic loading tests on an isolated truck rather than impulse loading or impact tests on a complete skateboard to eliminate deck characteristics and control for the positive strain-rate dependence exhibited in polyurethanes [4] and (2) remeasure a baseline bushing stiffness before each set of tests to permit offset correction of any batch effects.

This model notably leaves out the standard differentiating features between truck designs, such as the height of the hanger and the width of the axle that it holds.

However, the dimensions of these geometric elements, provided by the manufacturer, contribute to torque through constant scaling that can be trivially calculated as a ratio of the new moment arm to the old. Further reducing considerations between truck brands, the top and bottom cup washers around the bushings standardize the force application angle. With the experimental focus thoroughly isolated to the compliant bushings, a proper model by which to understand and characterize the bushings is required.

COMPLIANT BUSHINGS

The compliant bushing system takes two inputs of preload tightness, as measured by the torque on the kingpin in newton-meters, and the moment applied to the bushing, as measured by the force in newtons times the moment arm of the truck in meters, to an output of rotational displacement of the hanger in radians. The most apparent distinction between bushings is their geometry, categorized as barrels with constant outer diameter or cones with tapered outer diameter. While experiential wisdom dictates that barrel bushings prioritize stability and cone bushings emphasize responsiveness (a lack of damping in turns), publicly available data that controls for the variable of geometry are rare in consumer comparisons. From a theoretical perspective, geometry influences deformation characteristics through the cross-sectional area moment of inertia. Informal simulations have validated the distributors’ claims of tailored responsiveness in cone bushings [5], which appear to exhibit slight non-linearity in their initially weaker stress responses. However, with the characterization of Young’s modulus and standard beam bending equations, only the area moment of inertia must be calculated to predict how stress-strain curves change across geometries.

For further precision, the shear modulus for a given polyurethane formulation is necessary, but if experimentation is not possible, an estimated Poisson’s ratio of 0.5 can be reasonably extrapolated from similar elastomeric materials [6]. Therefore, the most significant variations in response from bushing geometry may be derived from a given bushing’s dimensions.

THE HARDNESS PROBLEM

The analysis of deformation presented above implies that the elastic modulus most directly describes a bushing’s response to load, but this interpretation must resolve its conflict with the industry standard of Shore-A hardness, or durometer. Hardness is a surface property that describes a material’s resistance to localized indentation [7], whereas

stiffness is a bulk property that describes the degree of deformation under a given load. Hardness only provides a rough correlation to stiffness, the property that designers and end-users are truly concerned with. Manufacturers' preference for the Shore A hardness scale over stiffness is not a quirk of the skateboard market but rather a ubiquitous standard of the elastomer industry, centered at the crossroads of many conveniences. While Young's modulus tests require expensive machinery and destructive measurements, Shore hardness can be determined with cheaper, handheld tools that permit quick, non-destructive quality testing [7]. Crucially, durometer serves as an intuitive proxy for a range of performance metrics by communicating the surface-level feel of an elastomer. This intuition applies to the skateboard bushings market due to the consumer understanding of 80A as a soft extreme (associated with maneuverability) and 100A as a hard extreme (associated with stability). This gap between hardness' convenience and stiffness' accurate description of behavior begs the academic question of how to translate between these correlated metrics.

The theoretical elasticity theory approach shown bridging international hardness standards in *Rubber Chemistry and Technology* produces Equation (1), where Young's modulus (E) in kg/mm^2 is expressed in terms of the unitless Shore hardness (s) and a proposed indenter radius (r) of 0.51mm [8]:

$$E = \frac{56 + 7.66s}{2.67r(254 - 2.54s)} \quad (1)$$

Gent's work was later directly validated in *Dental Materials*, where a set of simplifying assumptions on the material properties of elastomers, similar to those in this paper, were made to enable accurate stress calculations for soft materials in dental work [9]. However, dental work targets a Shore A hardness range of 30-60, so the study of harness values relevant to compliant skateboard bushings would expand the validation to the range of 80-100A.

Interrogating this 80-100A range, experiments comparing polyurethane elastomers of 80A and 90A hardness reveal that softer formulations plateau earlier in their stress-strain response, while harder ones maintain linearity through greater strain. These results underscore the positive correlation of stiffness and hardness along with the non-linearities that emerge from edge-case loading conditions, highlighting the need for the measurement and study of polyurethane's variable properties under realistic loading conditions [10].

POLYURETHANE

Polyurethane (PU), the material typically used for skateboard bushings, is a versatile elastomer with robust and tunable characteristics that allow for the application of polyurethane across industries [11]. Elastomers with hardnesses matching the 80-100A range of skateboard bushings often find use in vehicle suspension parts such as anti-roll bars or in construction and infrastructure rehabilitation [12]. Skateboard bushings represent a relatively small but high-performance niche in compliant parts where durability, tunability, and high yield strength converge.

Polyurethane's unique properties derive from its chemical structure, which holds the necessary foundation for analyzing and implementing elastomer systems. Synthesized as an amorphous, or uniformly distributed, polymer composed of entangled monomer chains, PU exhibits a dual-domain structure [13]. The soft domains make up a matrix of flexible polymer chains that can straighten under load and return to their highest entropy "tangled" state when the load is removed, inducing elasticity. The hard domains increase the Young's modulus, yield strength, and hardness through an ordered structure of chemical crosslinking. This chemical balancing act creates the "hyper-viscoelastic" behavior [14,15] that this project directly investigates, in which the material both elastically recovers from large strains (hyperelastic) and exhibits a time-dependent stress response (viscoelastic).

Viscoelasticity governs the unconventional behavior of PU that this paper quantifies. PU exhibits cyclic softening, creep deformation, positive strain-rate dependence [4,16], and most importantly, cyclic hysteresis, where loading followed by unloading causes energy dissipation. This discrepancy between the input and output work is typically represented by a "hysteresis loop," or the enclosed area on a stress-strain curve, as depicted in Figure 2, where the area under the loading (input) portion of the curve is less than the area under the unloading (output) portion.

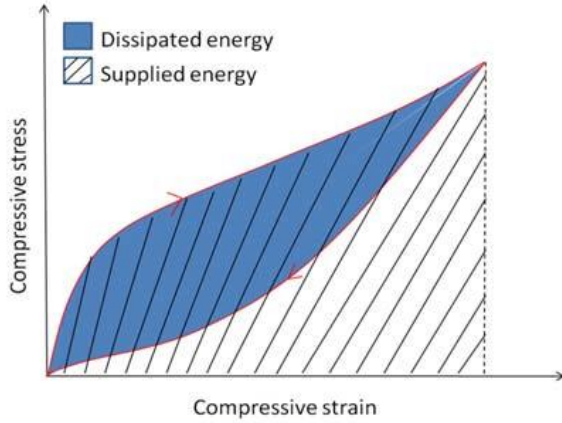


Figure 2: Generic hysteresis-loop on stress-strain graph with the supplied and dissipated energy clearly marked [17]. As PU, the material of skateboard bushings, is known to exhibit hysteresis, the dissipated energy is a relevant, measurable metric in its response under cyclic loads.

These phenomena are linked to the time-dependent, or “viscous,” motion of soft domains, which gradually flow into more energetically favorable conformations. The observation of significant hysteresis energy losses in PUs [18] could explain why stiffness, which is normally idealized as a constant quantity, varies with preload in skateboard bushings, as the preload bypasses a portion of the energy-intensive viscoelastic regime by partially untangling the soft domains.

Exploring the correlation between preload and stiffness at the macro level, stress-strain studies have found thermoplastic PUs, the specific classification of skateboard bushings, exhibit increased stress with increased amplitude of preconditioning [18,19]. This breaks the conventional understanding typically extrapolated from non-PU thermoplastics and rubbers, where more preconditioning causes material softening that reduces stress, known as the Mullins Effect [20]. Although the selected papers do not explicitly mention stiffness, their findings suggest that greater preloads can induce greater stress for the same strain when PUs are subjected to the extreme loading conditions relevant to skateboard bushings [19]. The increase in the ratio of stress to strain further supports the experiential hypothesis that preload positively correlates with stiffness in skateboard bushings. Together, these material insights contextualize the dual role of polyurethane bushings as both mechanical components and consumer-tunable elements, making their characterization central to understanding skateboard dynamics.

EXPERIMENTAL DESIGN

To test the deformation response of a bushing under realistic loading cycles, a Vernier Dual-Range Force Sensor applied a controlled load to a single fixed truck—set to a preload torque with a TRQ-21101 torque wrench—causing a gradual deflection in the truck that was measured by a LX-PA-10 linear transducer, as depicted in Figure 3. By analyzing the resulting force and displacement data within the context of the bushing-truck system’s geometry, the effective elastic moduli and hysteresis energy loss per trial were derived, plotted against preload torque, and fitted to physically informed power law relations.

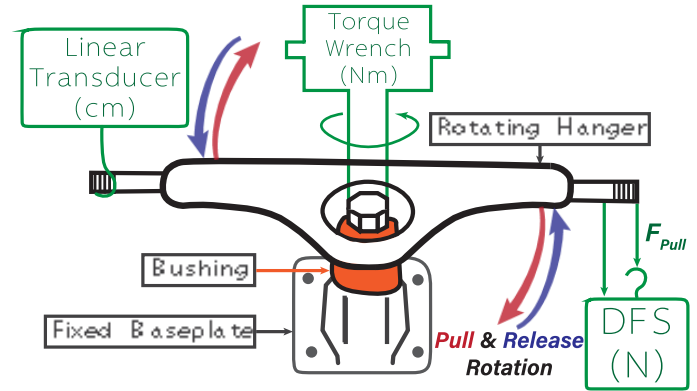


Figure 3: Diagram of testing apparatus around a single deflecting truck with a LX-PA-10 linear transducer, a Vernier Dual-Range Force Sensor (DFS), and a TRQ-21101 torque wrench that collectively obtain the data for a stress-strain curve associated with each trial’s preloading torque.

EXTRACTING STIFFNESS

The method of determining stiffness from each trial of preprocessed applied the following beam-bending approach. For each trial, displacement and force data were masked into loading and unloading phases, using displacement to identify the peaks rather than the force sensor that introduced noise as it neared the edge of its operating range. The radius of curvature (R) was derived from a small angle approximation with the known geometric parameters of the height of the bushing (h_0), the moment arm of the linear transducer (a), and the measured displacement (d), to obtain $R = (h_0 \times \frac{a}{d})$. Since the tested deflection range did not exceed 10° , the approximation of $\frac{d}{a}$ as bend angle in radians is accurate within 1%. With this finding, the formula for effective elastic modulus, Equation (2), was determined from three fundamental bending relations: $\sigma = \frac{M \times y}{I}$, $\epsilon = \frac{y}{R}$, and $E = \frac{\sigma}{\epsilon}$, where M is applied moment, y is distance from the neutral axis, and

I_x is cross-sectional area moment of inertia. The moment arm of the linear transducer and the force sensor were measured to be equivalent up to the precision of the analog calipers, so they are both notated as a .

$$E = \frac{M \times R}{I_x} = \frac{(F \times a) \times (h_0 \times \frac{a}{d})}{I_x} \quad (2)$$

The area moment of inertia corresponds to the dimensions of the tested bushings, which in this case corresponded to a frustrum of major radius R_2 and minor radius R_1 height h_0 , and a hole of radius r . From these dimensions, the area moment of inertia for a single cone bushing is provided separately in Equation 3, as different bushing geometries will necessarily call for different formulae.

$$I_x = \frac{\pi h_0}{20} (R_1^4 + R_1^3 R_2 + R_1^2 R_2^2 + R_1 R_2^3 + R_2^4 - 5r^4) \quad (3)$$

Equations (2) and (3) determine an effective elastic modulus for each force and displacement measurement, generating a wealth of data per trial. To analyze how stiffness varies with preload, each value was assigned the torque condition of its respective trial before being grouped into loading and unloading phases. To match the expected end behaviors of $0 \frac{\text{N}}{\text{m}}$ of stiffness with no preload and diminishing returns on stiffness as preload increased, power law fits were applied separately to the loading and unloading data, which are expected to demonstrate statistically significant difference due to polyurethane's viscoelasticity. These models performed better than the tested linear and exponential fits, which had larger 95% confidence intervals and non-uniform residual plots.

EXTRACTING ENERGY LOSS

The derivation of energy loss naturally proceeds from the force-displacement graphs, in a similar manner to the previously discussed Figure 2. Since the energy input and output are equivalent to the area under the loading and unloading phase respectively, the energy dissipated—visually the area inside of the hysteresis loop—can be calculated as the difference of these two integrals. This process returns energy lost in J per trial, which was plotted relative to each torque condition to reveal a negative correlation. Since energy dissipation must be positive and the loss is expected to asymptote as the bushing-truck system decouples when the preload torque approaches 0 Nm, a power law fit is most representative of the physical reality.

ISOLATED TRUCK SYSTEM

The design of a skateboard truck lends itself to isolating variables, as once the baseplate is fixed, the deflection of the hanger is directly fixed to the deformation of the bushing. However, constraints in instrumentation may limit a researcher's ability to measure along the unconventional, non-linear deflection paths the hanger takes. In these cases, inelastic string is recommended to rig sensors in lieu of a complex purpose-built apparatus. Such rigging resolved the range limitations of both the linear transducer and the force sensor in this setup. For the linear transducer, selected for its high analog precision, string simply extended the range of the LX-PA-10 beyond 10 inches so it could be positioned relative to the vice used to fix the truck's baseplate. For the Vernier Dual-Range Force Sensor, string was wrapped around the truck's axle, with one end attached to the force sensor and the other to a fixed point, creating a moveable pulley that effectively doubled the force sensors range from $\pm 50 \text{ N}$ to $\pm 100 \text{ N}$, as necessary to achieve the desired deflection, while halving its precision to a reasonable 0.1 N.

The TRQ-21101 torque wrench provides accuracy within 4% clockwise up to 17 Nm, which may be insufficient for harder bushings. For these cases, the alternate setup of a Vernier force sensor applying a perpendicular load to a skate tool (term for the T-shaped socket wrench multi-tool used to adjust skateboard parts) was validated to agree within the 4% error of the torque wrench. With this setup, applied torque is equivalent to the force sensor's maximum reading times the measured moment arm from the axis of rotation to the loading point. Crucially, both the linear transducer and force sensor were aligned to measure the typical roll axis deflection of the rotating hanger, as depicted in Figure 3. All relevant aspects of the geometry were measured with calipers, including the dimensions of the bushings and the moment arms from the force sensor and linear transducer to the center of the truck, which should both be roughly half of the truck's width to maximize mechanical advantage.

Force and displacement data were collected and preprocessed to minimize the effects of unmeasured variables within Logger Pro. To this end, each trial of 10 seconds evenly split into loading and unloading phases that peaked close to the same displacement of $\sim 0.4 \text{ cm}$, reducing variation in strain rate. To avoid significant contributions from cyclic softening, three unmeasured loading cycles were performed before nonconsecutive trials, as the literature suggests that the softening effects remain relatively constant after 2-3 cycles [20]. The raw force data was doubled to reflect the pulley's contribution

and the LX-PA-10's voltage data was converted to cm according to the datasheet's conversion factor. With particular attention to the removal of data after each trial so as not to consider residual strain as continued displacement, the preprocessed data was cleaned, formatted, and masked in MATLAB for the analysis detailed above.

RESULTS AND DISCUSSION

The quantitative relationships derived from the experimental data are presented in Figure 5 and Figure 6 and interpreted in the context of polyurethane's viscoelastic behavior. The following analyses address the two core objectives of this project: determining how kingpin preload influences the stiffness of the truck-bushing system and quantifying the corresponding energy lost to hysteresis during cyclic loading. These results provide a mechanistic understanding of how tightening a skateboard's trucks affects both its stability and responsiveness, forming the foundation for the accessible frameworks proposed in this work.

The representative force-displacement curve from a single trial displayed in Figure 4 illustrates the key behaviors discussed throughout this section. The loading (Pull) and unloading (Release) branches, shown in red and blue respectively, form a hysteresis loop where the enclosed area, highlighted in pink, represents the energy lost to hysteresis. The separation between the Pull and Release branches at 0 displacement indicates residual strain within the material, confirming the expected response for thermoplastic polyurethanes under cyclic loading [18]. These features confirm that the measured differences between loading and unloading are physically meaningful rather than experimental artifacts, validating the use of force-displacement data to characterize both elastic and dissipative properties of the system. The subsequent sections build on this representative behavior to quantify hysteresis energy loss, analyze the resulting trends in stiffness, and discuss the broader applications of these findings.

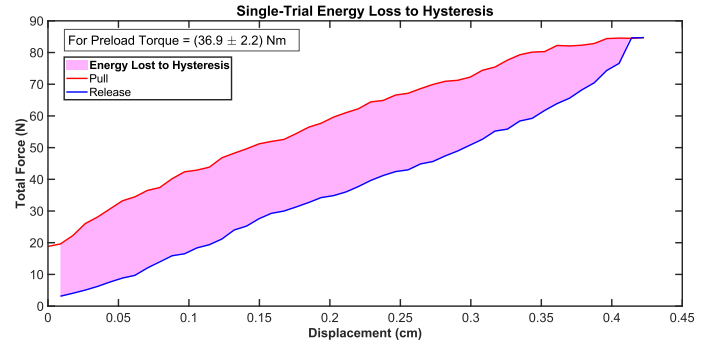


Figure 4: Representative single-trial plot of applied force (N) vs. truck displacement (cm) with loading (Pull) and unloading (Release) sections marked in red and blue respectively as well as energy lost to hysteresis (N×cm) highlighted. From this graph, the desired stiffness and energy loss data with the associated preload torque can be derived.

STIFFNESS

The objective of this portion of the analysis was to quantify how compressive preload from the tightening of the kingpin affects the stiffness of the truck-bushing system, as measured by its effective elastic modulus. By characterizing this relationship, a quantitative basis is established for how tightening the kingpin (a commonplace adjustment for skateboarders) translates into a predictable change in board stability and responsiveness. The stiffness analysis therefore connects directly to the project's central goal: relating user-controlled preload torque to measurable mechanical properties that can define and accessibly communicate turning performance.

The resulting stiffness data were fit to the power-law model, $E = aT^b$, where E represents the effective elastic modulus and T represents the applied torque to the kingpin. Separate fits for the loading (Pull) and unloading (Release) portions of the displacement cycle isolate the viscoelastic effects. The loading phases produced fit parameters of $a = (1.049 \pm 0.029) \times 10^4 \text{ m}^{-3}$ and $b = 0.4871 \pm 0.0087$, while the unloading phase produced $a = (7.73 \pm 0.21) \times 10^3 \text{ m}^{-3}$ and $b = 0.5114 \pm 0.0085$. The exponents cluster within $\sim 3\%$ of 0.5, suggesting an approximately square-root relationship between preload torque and stiffness. Across the tested torque range, stiffness increased by $(97 \pm 5)\%$, confirming a strong, nonlinear dependence between the two parameters. The 95% confidence intervals of the fitted curves, shown as shaded regions in Figure 3, are clearly distinct between the Pull and Release data, indicating statistically significant hysteresis behavior.

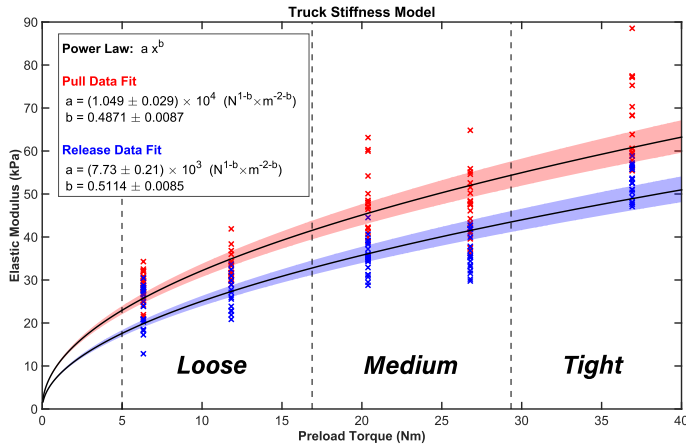


Figure 5: Stiffness of the truck system (kPa) vs. preloading torque applied to kingpin (Nm), separated by loading (pull) and unloading (release) data with respective power law fits and 95% confidence intervals highlighted, quantifies how stability, proportional to the stiffness of the truck system, depends on the user-controlled kingpin bolt preload. To contextualize the relationship in experiential vocabulary, the torque range is divided into three zones, Loose, Medium, and Tight, by dashed vertical lines.

To contextualize the findings, the torque range in Figure 1 is divided into three zones, Loose, Medium, and Tight, by dashed vertical lines. These zones provide a mapping between experimental torque values and the vocabulary skateboarders use to describe board feel. Such zones represent only the beginning of this model's accessibility to end-users, as the utility of stiffness enables the streamlined analysis and creation of frameworks to communicate tacit understandings of bushings from the surplus of known variables surrounding truck dimensions and user preference, as previously outlined.

The square root trend identified in both loading and unloading data suggests a physically interpretable relationship between preload and stiffness. The exponent may emerge from the combined effects of geometric and material nonlinearities. As preload increases, the effective lever arm shortens and the bushing's stress distribution shifts, producing sub-linear stiffness growth in a manner that the small-angle approximations in the geometry analysis could not capture. The positive, non-linear correlation between preload and stiffness complements the literature describing thermoplastic polyurethane's deviation from the Mullins effect, in which higher preconditioning typically softens rubbers and thermoplastics [20]. Instead, the observed trend indicates that preloading in the range relevant to skateboard

bushings enhances the material's ability to resist deformation, consistent with the hypothesis of stress-dependent rearrangement of the underlying viscoelastic chemical structure in thermoplastic polyurethanes [19].

The statistically significant offset between the Pull and Release stiffness values, clearly present in the non-overlapping 95% confidence bands in Figure 5, further supports the presence of viscoelastic hysteresis. The $(21 \pm 12)\%$ difference in effective modulus between loading and unloading corresponds to the energy-intensive restructuring of soft domains for deformation, matching theoretical expectations for hyper-viscoelastic materials. This distinction between loading and unloading behavior provides a physical explanation for the presence of the subjective damping skateboarders experience in turns, as the energy lost to hysteresis is experienced as an incomplete force transfer through the bushing.

ENERGY LOSS

The next experimental goal was to quantify how the mechanical energy the truck-bushing system dissipates during a loading cycle depends on kingpin preload. The hysteresis data in Figure 6 were well described by a decaying power law $E_{loss} = aT^b$ with fitted parameters $a = 0.364 \pm 0.096 (Nm)^{1-b}$ and $b = -0.35 \pm 0.10$. The statistically significant negative exponent indicates a robust decrease in energy loss as preload increases over the tested range relevant to skateboard bushings. Practically, this fit validates that tightening the kingpin reduces the energy dissipated per cycle, as suggested by the trend of residual strain decreasing and shrinking hysteresis loops in the force-displacement plots of higher preload conditions. The 95% confidence band of Figure 6 demonstrates that the decrease is reliable yet gradual across the measured torque conditions.

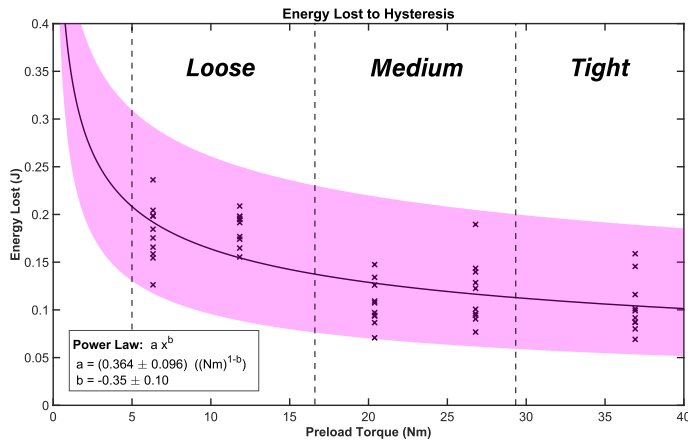


Figure 6: Energy lost to hysteresis (J) vs. preloading torque applied to kingpin (Nm), with a power law fit and 95% confidence intervals highlighted, quantifies how responsiveness, inversely proportional to the energy lost per trial of cyclic loading, depends on the user-controlled kingpin bolt preload. To contextualize the relationship in experiential vocabulary, the torque range is divided into three zones, Loose, Medium, and Tight, by dashed vertical lines.

These observations tie directly to the viscoelastic properties of skateboard bushings. Increasing preload alters the underlying material chemistry of polyurethane, partially untangling soft domains and reducing the portion of deformation that is impeded by the dissipative molecular rearrangement [19]. Effectively skipping the most energy-intensive portion of viscoelastic deformation, increased preload causes a chemical restructuring that translates into a macroscopic reduction in damping that is apparent to the user. This model, therefore, provides a direct, quantitative mapping from user input to user experience, by quantifying how a skateboarder's torque adjustment can be used to control the damping felt during turns.

APPLICATIONS AND FUTURE WORK

The stiffness and energy-loss findings form an actionable takeaway: kingpin preload is a user-accessible control that predictably increases stiffness and reduces hysteresis. This dual effect directly contributes to the project's motivation of democratizing information surrounding skateboard tuning. By translating torque into mechanically meaningful quantities such as elastic modulus and joules lost per cycle, the results enable experiential expertise to be converted into accessible quantitative recommendations. The mappings offer a refined basis for specifying target performance for

manufacturers and development engineers, while providing a clear path to simple, reproducible guidance for skateboarders and shops, such as clear tables that identify how much torque a rider in a given weight range should apply for a desired board feel (loose, medium, or tight).

While the results demonstrate strong, consistent torque-stiffness and torque-hysteresis relationships, limitations remain in the models' accuracy and generality. The analysis presently reflects a single bushing geometry and hardness, with trials split across five torque conditions. The concentration of trials at five torque conditions introduces uncertainty in the fit coefficients, while complicating further analysis that could isolate secondary factors, such as rate dependence, from torque. The applied beam-bending model neglects shear deformation and rate dependence, both of which may bias the magnitude of the elastic modulus while preserving its general trend. With specific respect to the energy loss data, instrument noise at the extremes of each hysteresis loop along with the masking choices made to identify them influence how energy is calculated. Taken together, these findings call for the testing of additional bushing geometries and formulations to determine whether the square-root relationship persists, to validate hardness-to-stiffness relationships in the literature [8], and to characterize strain-rate effects [4] explicitly. Such extensions would strengthen the generality of the model and enhance its utility for both manufacturers and users seeking insight into tuning their individualized skateboard setup.

This suggested line of work will produce clear translational opportunities. The discussed short-term applications include lookup tables relative to known quantities such as rider weight and truck height, torque-recommendation stickers to place on trucks for new users, quick guides for skate shops, and a validated procedure for measuring bushing performance in small labs and shops. Long-term outcomes include informing the design of low hysteresis bushing formulations, incorporating quantitative tuning curves into toolkits used by professional coaches and manufacturers, and the rethinking of how trucks can communicate the relationship of torque to board feel so that new and experienced users alike no longer struggle with blind trial and error.

Although broader validation across many bushings and usage conditions remains necessary, the present results provide a principled foundation for democratizing bushing knowledge, helping skateboarders, shops, and designers move from subjective feel to reproducible settings and universally accessible communication.

CONCLUSIONS

This study quantitatively established the relationship between kingpin preload torque and the mechanical behavior of skateboard bushings, providing a rigorous framework for translating reproducible user adjustments into measurable mechanical effects. Across the tested loading conditions, stiffness increased approximately with the square root of torque, following the applicable empirical relation of $E = aT^{0.5}$, where quadrupling torque doubles stiffness. Stiffness rose by $(97 \pm 5)\%$ over the studied range of preload torques, which are representative of a typical bushing's operating range, confirming a strong, nonlinear dependence between preload and elastic modulus. Across the same range, hysteresis energy loss followed a decaying power-law while remaining statistically significant, which quantifies the effects of viscoelastic properties with validation from the statistically significant difference between loading and unloading effective elastic moduli. These results quantify qualitative skateboarding wisdom: tightening the kingpin simultaneously enhances stability and reduces damping, while looser settings favor maneuverability at the expense of greater energy dissipation.

The implications of these findings extend beyond the single isolated truck. By relating mechanical preload to measurable stiffness and energy loss, this work bridges the gap between subjective board feel and objective and useful engineering parameters. The derived models directly inform the translation of reproducible torque measurements into stiffness and damping values that dually guide both design and user adjustment. In practical terms, these models establish the foundation for clear communication and streamlined analysis, such as standardized tuning charts that enable skateboarders to select torque settings based on rider weight and truck geometry rather than relying on one's proficiency in experiential trial and error. Manufacturers and local skate shops can further apply these mappings to communicate part recommendations with clarity and consistency, lowering barriers for beginners and empowering independent businesses with accessible mechanical knowledge.

Beyond their immediate applications in the skateboarding industry, these results contribute to the broader understanding of viscoelastic polyurethane systems under compressive preloads. The observed enhancement in stiffness and reduction in hysteresis with increasing torque confirm the theoretical expectation that the preloading of thermoplastic polyurethanes shifts deformation into a less energy-intensive regime [19]. This

behavior particularly highlights polyurethane's unique hyper-viscoelastic properties due to its contrast with the Mullins softening typically observed in rubbers and thermoplastics [20]. The methodology demonstrated in this study, where compliant mechanisms are isolated and characterized through constant torque-controlled preloads, offers a transferable framework for analyzing other elastomeric components where tunability plays a major role.

While this study was limited to a single bushing geometry and hardness, the consistency and statistical significance of the derived relationships suggest that similar patterns may extend across other formulations. Future work should test more relevant geometries, hardness ranges, and dynamic loading rates to refine the model and assess the effects of strain-rate. Such efforts would enable precise correlations between durometer, stiffness, and perceptible board feel, while expanding the utility of hardness studies [8, 10] to end users. Even with these considerations, the present results provide a principled, experimentally validated pathway toward further democratization of tacit skateboard equipment knowledge, transforming elusive intuition into accessible, quantitative understanding. In doing so, this work not only advances the scientific mechanics of skateboarding but also strengthens the community foundations that will sustain its continued growth.

ACKNOWLEDGMENTS

The author thanks Dr. Hughey for her insight into experimental design, Dr. Lachner for his great influence on the focus and technical communication of this study, and Mr. Braunstein for his guidance in instrumentation.

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